FermatQuinticHost: Init-Only Lean 4.22 Host Certificate for the Fermat Quintic Fourfold V(F_5) in P^5

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/FermatQuinticHost
Dedicated 1-to-1 Zenodo DOI: 10.5281/zenodo.23652990

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Abstract

Standalone Init-only Lean 4.22 formalization (‘Hodge/VF5.lean‘) defining the host arithmetic and degree/weight invariants of the smooth Fermat quintic fourfold V(F_5) subset P^5 with zero external Mathlib dependencies.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Explicitly certified as a host definition and arithmetic invariant module (‘Hodge/VF5.lean‘) with Z uncomputed; not a proof or disproof of the Clay Hodge Conjecture.

- **Repository:** `https://github.com/BenFrohman/FermatQuinticHost`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652990`
- **Earliest Git Commit SHA:** `799c9815446e3e64d7f21701d0efdfdbed697ae5` (2026-10-03 04:45:56 PM CDT (UTC-5))
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2 Technical Overview & Repository Specification

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The real host is the Fermat quintic fourfold
“‘text
F5 = x0^5 + x1^5 + x2^5 + x3^5 + x4^5 + x5^5
V(F5) subset P^5
“‘
Certificate: [‘Hodge/VF5.lean‘](Hodge/VF5.lean) and [‘CERTIFICATE.md‘](CERTIFICATE.md).
‘Hodge/NewHost.lean‘ is a placeholder filename, not the host name.
‘hostLabel‘ is a string. ‘T_F‘ is not used. ‘Z‘ on the sextic is uncomputed.
Primitive ‘H^{3,1}‘ container dimension is 120, not 0.
Toolchain: Lean ‘v4.22.0‘, Init only.

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/FermatQuinticHost` is audited via `titan-audit` (`audit/TITAN_AUDIT_CE` and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
`CERTIFICATE.md`	873 B	`6d0078b674735413...`
`Hodge/VF5.lean`	4476 B	`e69d4e4a737ddb28...`
`LICENSE`	11355 B	`9da72c2bee4f20ba...`
`NOTICE`	329 B	`b4790083f7f73b66...`
`README.md`	531 B	`e2706b1cc1d2a2fa...`
`lean-toolchain`	25 B	`8f6680f8389e8ade...`

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